EED 101 INTRODUCTION TO ELECTRICAL ENGINEERING

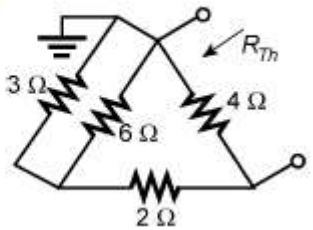
SPRING 2017 **INSTRUCTOR: DR G NAVEEN BABU**

PROBLEM SET 5

REFERENCE TEXT BOOK: INTRODUCTORY CIRCUIT ANALYSIS, ROBERT BOYLESTAD, ***12TH EDITION***

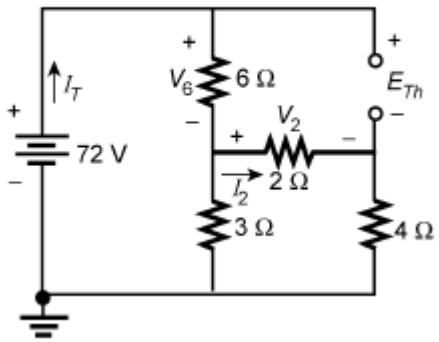
Chapter 9: 13,14,17, 26,27,29

13. R_{Th}:



$$R_{Th} = 4\ \Omega \parallel (2\ \Omega + 6\ \Omega \parallel 3\ \Omega) = \mathbf{2\ \Omega}$$

E_{Th}:

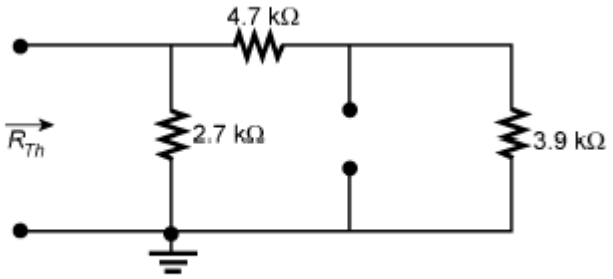


$$I_T = \frac{72\ \text{V}}{6\ \Omega + 3\ \Omega \parallel (2\ \Omega + 4\ \Omega)} = 9\ \text{A}$$

$$I_2 = \frac{3\ \Omega(I_T)}{3\ \Omega + 6\ \Omega} = \frac{3\ \Omega(9\ \text{A})}{9\ \Omega} = 3\ \text{A}$$

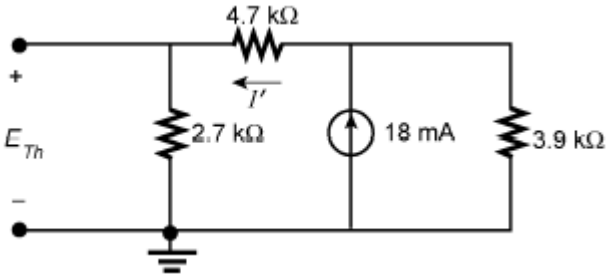
$$E_{Th} = V_6 + V_2 = (I_T)(6\ \Omega) + I_2(2\ \Omega)$$
$$= (9\ \text{A})(6\ \Omega) + (3\ \text{A})(2\ \Omega) = \mathbf{60\ V}$$

14. a. R_{Th}:



$$R_{Th} = 2.7\ \text{k}\Omega \parallel (4.7\ \text{k}\Omega + 3.9\ \text{k}\Omega) = 2.7\ \text{k}\Omega \parallel 8.6\ \text{k}\Omega = \mathbf{2.06\ k\Omega}$$

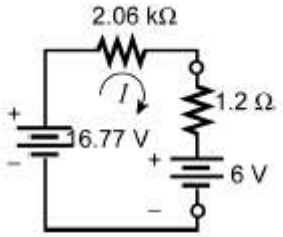
E_{Th}:



$$I' = \frac{3.9\text{ k}\Omega(18\text{ mA})}{3.9\text{ k}\Omega + 7.4\text{ k}\Omega} = 6.21\text{ mA}$$

$E_{Th} = I'(2.7\text{ k}\Omega) = (6.21\text{ mA})(2.7\text{ k}\Omega) = \mathbf{16.77\ V}$

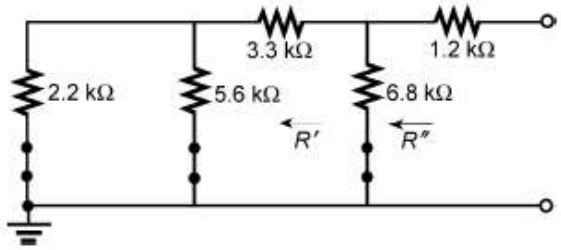
b.



$$I = \frac{16.77\text{ V}}{2.06\text{ k}\Omega + 1.2\text{ k}\Omega} = \frac{16.77\text{ V}}{3.26\text{ k}\Omega}$$

$= \mathbf{5.14\ mA}$

17. R_{Th}:



$\leftarrow R_{Th} = 2.2\text{ k}\Omega \parallel 5.6\text{ k}\Omega = 1.58\text{ k}\Omega$

$R' = 1.58\text{ k}\Omega + 3.3\text{ k}\Omega$

$= 4.88\text{ k}\Omega$

$R'' = 4.88\text{ k}\Omega \parallel 6.8\text{ k}\Omega = 2.84\text{ k}\Omega$

$R_{Th} = 1.2\text{ k}\Omega + R'' = 1.2\text{ k}\Omega + 2.84\text{ k}\Omega = \mathbf{4.04\ k\Omega}$

E_{Th}: Source conversions:

$$I_1 = \frac{22\text{ V}}{2.2\text{ k}\Omega} = 10\text{ mA},\ R_s = 2.2\text{ k}\Omega$$

$$I_2 = \frac{12\text{ V}}{5.6\text{ k}\Omega} = 2.14\text{ mA},\ R_s = 5.6\text{ k}\Omega$$

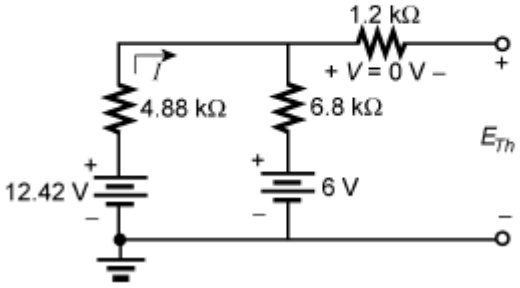
Combining parallel current sources: $I'_T = I_1 - I_2 = 10\text{ mA} - 2.14\text{ mA} = 7.86\text{ mA}$

$$2.2\text{ k}\Omega \parallel 5.6\text{ k}\Omega = 1.58\text{ k}\Omega$$

Source conversion:

$E = (7.86\text{ mA})(1.58\text{ k}\Omega) = 12.42\text{ V}$

$R' = R_s + 3.3\text{ k}\Omega = 1.58\text{ k}\Omega + 3.3\text{ k}\Omega = 4.88\text{ k}\Omega$

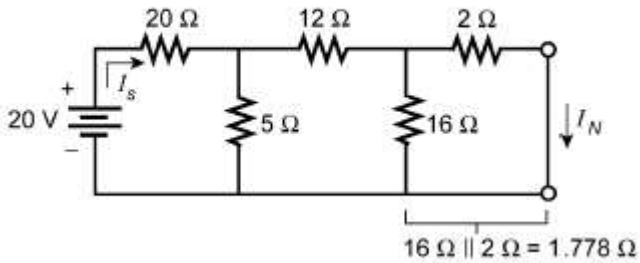


$$I = \frac{12.42\ \text{V} - 6\ \text{V}}{4.88\ \text{k}\Omega + 6.8\ \text{k}\Omega} = \frac{6.42\ \text{V}}{11.68\ \text{k}\Omega} = 549.66\ \mu\text{A}$$

$$V_{6.8\text{k}\Omega} = I(6.8\ \text{k}\Omega) = (549.66\ \mu\text{A})(6.8\ \text{k}\Omega) = 3.74\ \text{V}$$

$$E_{Th} = 6\ \text{V} + V_{6.8\text{k}\Omega} = 6\ \text{V} + 3.74\ \text{V} = \mathbf{9.74\ V}$$

26. From Problem 15, $R_N = R_{Th} = \mathbf{10\ \Omega}$



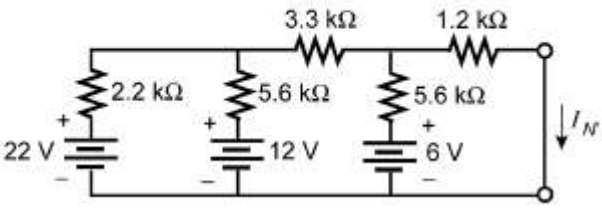
$$R_T = 20\ \Omega + 5\ \Omega \parallel (12\ \Omega + 1.778\ \Omega) = 23.67\ \Omega$$

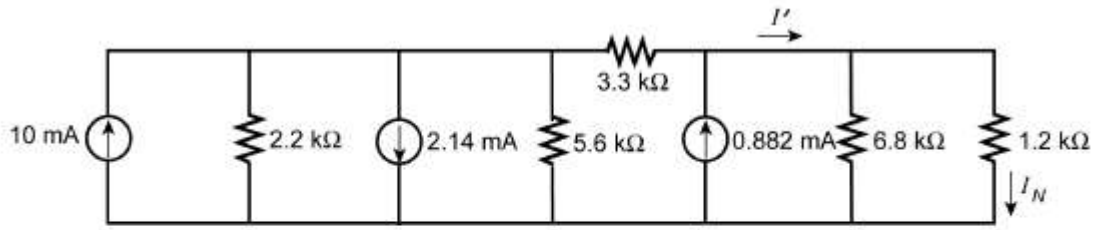
$$I_s = \frac{E}{R_T} = \frac{20\ \text{V}}{23.67\ \Omega} = 844.95\ \text{mA}$$

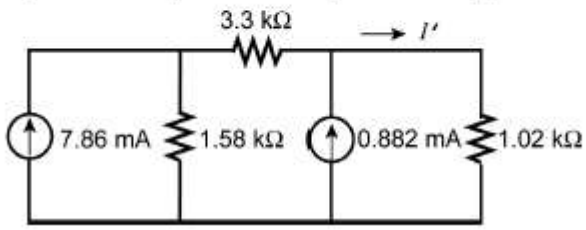
$$I_{12\Omega} = \frac{5\ \Omega(844.95\ \text{mA})}{5\ \Omega + (12\ \Omega + 1.778\ \Omega)} = 224.98\ \text{mA}$$

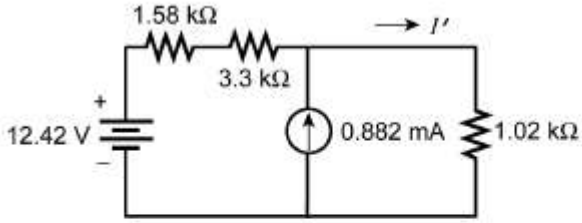
$$I_N = \frac{16\ \Omega(224.98\ \text{mA})}{16\ \Omega + 2\ \Omega} = \mathbf{200\ mA}$$

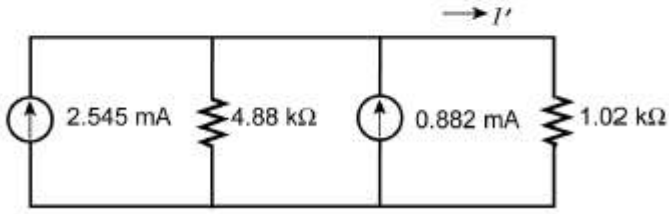
27. From Problem 17, $R_N = R_{Th} = \mathbf{4.04\ k\Omega}$

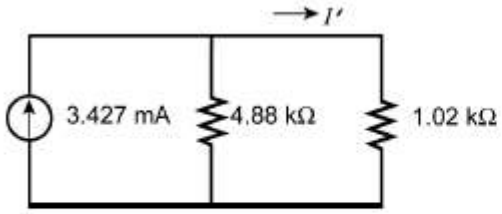








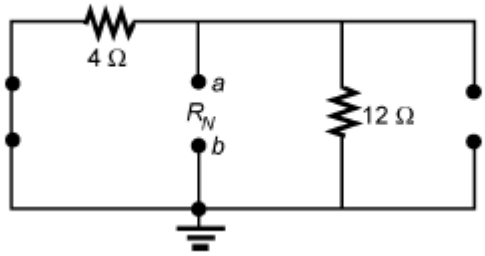




$$I' = \frac{4.88\ \text{k}\Omega(3.427\ \text{mA})}{4.88\ \text{k}\Omega + 1.02\ \text{k}\Omega} = 2.83\ \text{mA}$$

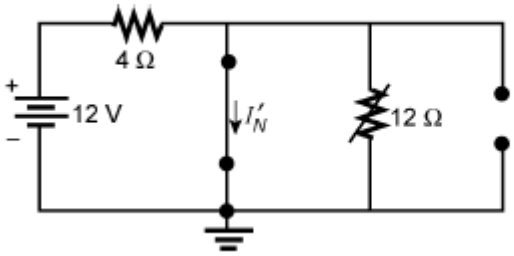
$$I_N = \frac{6.8\ \text{k}\Omega(2.83\ \text{mA})}{6.8\ \text{k}\Omega + 1.2\ \text{k}\Omega} = \mathbf{2.41\ mA}$$

29. a. R_N:



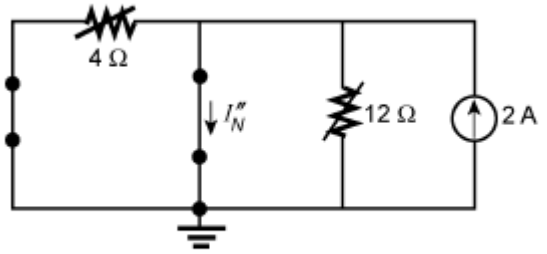
$$R_N = 4\ \Omega \parallel 12\ \Omega = \mathbf{3\ \Omega}$$

$E = 12$ V:



$$I'_N = \frac{12\ \text{V}}{4\ \Omega} = 3\ \text{A}$$

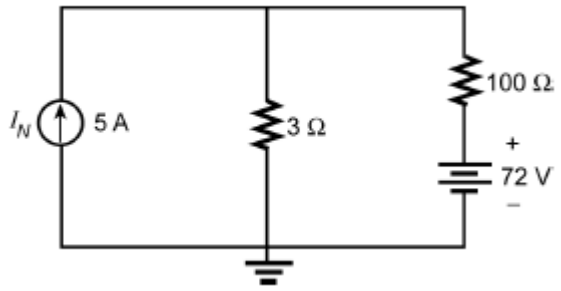
$I = 2$ A:



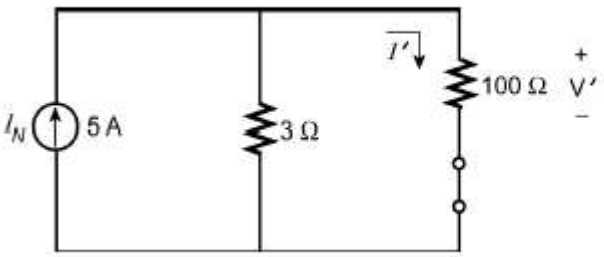
$I''_N = 2$ A

$I_N = I'_N + I''_N = 3\text{ A} + 2\text{ A} = \mathbf{5\ A}$

b.



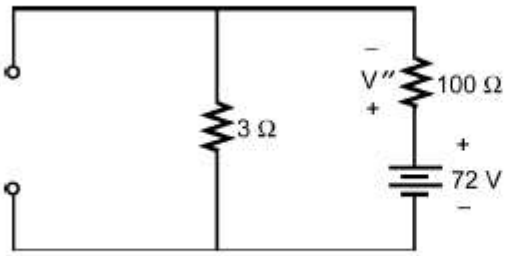
I:



$$I' = \frac{3\ \Omega(5\text{A})}{3\ \Omega + 100\ \Omega} = 145.63\text{ mA}$$

$$V' = IR = (145.63\text{ mA})(100\ \Omega) = 14.56\text{ V}$$

E:



$$V'' = \frac{100\ \Omega(72\text{ V})}{100\ \Omega + 3\ \Omega} = 69.9\text{ V}$$

$$V_{100\Omega}{}_{+}^{-} = V'' - V' = 69.9\text{ V} - 14.56\text{ V} = \mathbf{55.34\ V}$$